# Characterize Before You Loop: What Transfers From a Zero-Shot Proposer's Template and What Does Not

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## Abstract

A weak-tier language model proposer, asked in a zero-shot call to pack circles in the unit square, concentrates on a grid-plus-filler template. A companion submission gives the closed form naming its modal output. That template is the anchor, where its outputs settle at a given size. Its recipe family has a largest member there, the family argmax. This paper measures which survives a change of setting. Eight registered arms change four things. They vary the parent, the iteration, the serving path with its reasoning setting, and the vendor. Under a better in-family parent, on the agent-runtime instrument, 26 of 26 valid outputs return that parent's score, not the anchor. Under its own anchor as parent the model leaves that anchor, missing the registered bar. Iteration erodes it across five archive-and-mutate generations on that instrument. The same prompts on a pinned path return 0 of 105 valid packings. Extended thinking restores validity at the one cell tested. At a second vendor the pooled bar is met while the cells that test anchoring miss it. Under an instruction to improve, the anchor does not persist. The family argmax does, as a ceiling no arm clears at $10^{-9}$. Clearing it would mean leaving the recipe family, and no valid output did.

## 1 Introduction

Ask a language model to place 21 circles in a unit square to maximize the sum of their radii. It will not search. It reaches for the nearest square grid, five by five ($k = 5$) at radius $1/(2k)$, and truncates it, leaving four cells empty. A companion submission (Anonymous authors, 2026) gives that behaviour a closed form, naming the modal output in advance. It reports that form as the empirical mode at all seven $N$ tested, at four of the five held out, and under a perturbed container and paraphrase. The two cell sets are listed in Table 2. This paper's own generation-0 rows were sampled fresh under its loop arm, on the same instrument and prompt bytes (Anonymous authors, 2026). At $N = 31$ they put the anchor on 7 of 8 valid outputs. At $N = 13$, on 4 of 9 (Table 1).

A characterization that sharp invites an inference. What a proposer emits zero-shot is what it will contribute inside a discovery loop, so the loop need not run. Discovery systems in the FunSearch line (Romera-Paredes et al., 2024; Novikov et al., 2025) do not call their proposer zero-shot. They condition it on a parent and iterate it against an archive. Each is a change of setting, as is the serving path (Anonymous authors, 2026). Each can break the characterization.

**Objects and arms.** A *template* is one member of the grid-plus-filler recipe family. The *anchor* is the template the model's outputs concentrate on at a given $N$. The *family argmax* is the largest value the family can express at that $N$. A *discriminating cell* is an $N$ where the two differ, so only there can anchoring be told from family search. Table 2 carries the rest. Eight registered arms change four things. Arm MU changes the parent. Arm L adds iteration, feeding a scored proposal back as the next parent. Arms P and P-D change the serving path and the reasoning setting on it. Arms GM, GM2, GM3 and V change the vendor, GM2 and GM3 differing only by output budget.

**Claim.** A weak-tier LLM proposer's zero-shot anchor in circle packing is contingent on the serving path and its request parameters. A zero-shot characterization therefore licenses no prediction inside a discovery loop, absent a measurement there. The anchor does not transfer. The family argmax does, as a ceiling. No valid output clears that ceiling in any arm at $10^{-9}$, and at the second vendor's discriminating cells most valid outputs sit on it. The primary $10^{-6}$ tolerance admits circle overlaps up to that size. Under it, 18 arm MU outputs sit above the argmax by at most $1.53 \times 10^{-6}$, an excess that overlap accounts for. All 18 are valid at $10^{-6}$ and fail at $10^{-9}$ (§4). Two arms carry no row in Table 1. Arm GM finished UNDERPOWERED whether its routed rows are counted or not. Arm GM2 returned nothing parseable at the smaller output budget. Arm GM3 diagnosed the cause at the larger.

Table 1: Six of the eight arms: what each changed, its bar, the outcome and the verdict. Codes like P-MU1 name a registered prediction, written out in its arm's section and Supplement S1. L1 to L4 are the loop arm's four, named in §5. Arms MU and L ran on the agent-runtime instrument, P and P-D on the pinned path. Bracketed ranges are 95% Wilson intervals on the rate they follow.

| Arm | What changed | Registered bar | Outcome | Verdict |
|---|---|---|---|---|
| MU | The parent the proposer is conditioned on | Under a better in-family parent, on-prediction rate at most 20% and keep-or-improve at least 50% | 26 of 26 keep or improve [87–100%], 25 within $10^{-6}$ of the parent's score and one $1.2 \times 10^{-6}$ above it, so the on-prediction rate there is 0 of 26 [0–13%]; under the model's own anchor, 7 of 28 keep it [13–43%] and 17 move above it | The rule reduces to one condition under the better parent; the own-anchor rate misses P-MU1's 50% bar, so the anchor is not self-stable; the falsifier is that rule's negation and adds no second result |
| L | Five archive-and-mutate generations | Pooled conditioned on-prediction rate below the generation-0 rate in both cells (L2); the falsifier is its negation | $N = 13$: 4 of 9 at generation 0, 5 of 20 conditioned. $N = 31$: 7 of 8, then 6 of 29. Pooled 11 of 17 [41–83%], then 11 of 49 [13–36%] | L2 holds; L1 and L3 not met; L4 open (0 of 49 clear the family, [0–7%]) |
| P | The serving path: the bare pinned chat-completions endpoint | A floor of five valid outputs per cell | 0 of 105 valid at the square cells [0–4%]; 4 of 75 at the held-out cells [2–13%] | Every direct-emission prediction UNSCOREABLE: no cell reaches the floor |
| P-D | Extended thinking on against off, same pinned path | The reasoning budget is the cause, against the missing context, against anything else | Thinking on, 12 of 14 valid with the anchor the plurality value at 5 of 12 [19–68%]; off, 0 of 13 [0–23%] | The reasoning-budget hypothesis holds; extended thinking is the component |
| GM3 | A second vendor's gemma at a 16,384-token budget | Pooled rate at least 30%, and the anchor modal in 5 of 5 scoreable cells | 46 of 80 pooled (57.5%, [47–68%]); modal in 2 of 5; 11 of 43 at the discriminating cells (26%, [15–40%]) | The pooled bar is met; the discriminating cells, the only ones that test anchoring, miss it |
| V | Thirteen free-tier aliases | At least 10 of a model's 25 invocations valid, then V1 at least 50% and V2 at most 10% | Ten aliases below the floor; two aliases meet the V1 pooled bar at point estimate, on 8 of 12 [39–86%] and 11 of 18 [39–80%] of their valid outputs, and a third does not, at 0 of 15 [0–20%] | The boundary is located, not attributed |

**Contributions.**

1. *A measurement of transfer, not an argument about it.* Each change of setting got its own arm, with thresholds and a falsifier fixed before the first invocation. Every deviation is tabled.
2. *A named instrument component, at one cell.* On the pinned path at $N = 13$, arm P's validity collapse is not caused by a missing system prompt. Adding a system line leaves it in place. One registered

condition (P-D, whose rows both submissions report) and two post hoc diagnostics separate the candidates. Extended thinking carries the anchor.

3. *A cross-vendor boundary that is located but not attributed.* Two second-vendor measurements disagree, with model variant and instrument confounded.

**Scope.** The proposer is one vendor's weak-tier alias (`claude-haiku-4.5`) except where another is named. The conditioning arm runs at three trap cells, $N = 13$, 21 and 31. The loop arm runs at two of them.

*What is not claimed.* Nothing about the companion paper's numerical baseline, about which constructions beat the family argmax, or about its mid-tier alias. Those are its results, cited and not restated (Anonymous authors, 2026). Nothing about mechanism inside the weights, about what a discovery system does at generation zero, or whether diversity machinery (Mouret & Clune, 2015; Lehman & Stanley, 2011) repairs or explores. Preregistration is an adopted standard, not a contribution (§8).

## 2 The template and its closed form

This section restates the object under test. Every definition below is the companion paper's (Anonymous authors, 2026). Only the re-score at the end is this paper's own.

**The benchmark.** Maximize the sum of radii of $N$ non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates, so scoring needs no human. The direct-emission prompt fixes the count, states containment and non-overlap, forbids code, and demands a raw Python list of `[x, y, r]` triples. The program channels replace that clause with a program contract. The prompt and its digest are in the supplement.

**The recipe family.** The **recipe family** is the parametric set of grid-plus-filler constructions. A **template** is one member of it. The **anchor** is the template the distribution concentrates on at a given $N$.

A $k \times k$ grid places one circle per cell center with $r_{\text{grid}} = 1/(2k)$. Then $m$ fillers sit on interior grid vertices, tangent to the four grid circles around them, with $r_{\text{filler}} = (\sqrt{2} - 1)/(2k)$ and $0 \le m \le (k-1)^2$. Hence $V(k, m) = k/2 + m(\sqrt{2} - 1)/(2k)$.

When $N < k^2$ a smaller complete grid is available, the $(k-1) \times (k-1)$ lattice at radius $1/(2(k-1))$ with fillers on its interior vertices. The model does not fall back to it. It *truncates* instead, laying out the $k \times k$ lattice, occupying $N$ cells and keeping the radius, which gives $T(k, N) = N/(2k)$.

**The order rule and the branch rule.** Write $k^*(N) = \text{round}(\sqrt{N})$ for the nearest-square order. This is a definition, not a hypothesis. The falsifiable content is the **branch rule**. Where $k^{*2} \le N$ the model extends, adding $m = N - k^{*2}$ fillers: the converge branch. Where $k^{*2} > N$ it truncates the $k^*$-grid: the trap branch. Substituting $k^*$ into the branch condition gives the *trap zones* $N \in [k^2 - k + 1, k^2 - 1]$, which over the sweep range are $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$ and $[57, 63]$.

The **family argmax** at $N$ is the largest value the family can express there, recomputed in closed form over every admissible $(k, m)$. Inside a trap zone it is the larger of two candidates: the drop-and-fill $V(k^*{-}1, N - (k^*{-}1)^2)$ and the truncated anchor $T(k^*, N) = N/(2k^*)$. At $N = 35$ the truncated anchor is larger, so 35 is not a discriminating cell, an $N$ where the anchor and the family argmax differ. At a converge cell the extended anchor is the argmax, so every discriminating cell is a trap cell. The anchor there is the truncate branch $T(k^*, N) = N/(2k^*)$.

**Independent re-score.** A second implementation recomputed every count in this paper (`arm_transfer_independent_rescore.py` in the artifact). The authors wrote it for this submission, with its own parser, its own validity scorer and a linear program blind to the recipe. It imports nothing from the arms' scoring code. It re-derived the closed-form value and the family argmax at all twelve $N$, agreeing to at most $3.6 \times 10^{-15}$. Of 135 cell-level counts across all arms, 129 agree. The six that differ are three arm L clearance counts at $N = 31$, two arm GM validity counts and arm P-D's sampled-row count, each with its cause in Supplement S1. The re-score also found an arm MU bucketing error, logged as item 36 of the artifact's corrections ledger, and §4 carries the corrected figures.

**Notation and conventions.** Table 2 restates the companion paper's terms.

Table 2: Terms used throughout this paper, each used in this sense only.

| Term | Meaning |
|---|---|
| on-prediction | emitted sum within $2 \times 10^{-3}$ of the anchor |
| family argmax | the highest-valued member of the recipe family at that $N$: inside a trap zone, the drop-and-fill $V(k^*-1, m)$ wherever it beats the truncated anchor. This paper gives it no other name |
| square, held-out and code cell | the three cell sets: a *square cell* is an $N$ addressed with the bare direct-emission prompt, $N = 13$, 17, 21, 31, 35, 37, 43; a *held-out cell* is an $N$ the companion paper kept out of its fit and this paper reuses, $N = 50$, 58, 62, 65, 75; a *code cell* is an $N$ addressed under the program contract instead, $N = 13$, 21, 31 |
| discriminating cell | $N$ where anchor $\neq$ family argmax; only these can distinguish anchoring from family search |
| validity | non-overlap and containment at $10^{-6}$ (primary) and $10^{-9}$ (logged), both always reported |
| trap cell | an $N$ inside a trap zone, where the branch rule truncates; the discriminating cells are the trap cells whose truncated anchor sits below the family argmax |
| tier | nominal capability class of the serving alias addressed: here the weak tier is the vendor's Haiku alias (`claude-haiku-4.5`) |
| instrument | the proposer's serving path with the request parameters it carries, the reasoning setting included. Two are used here, an agent runtime and a pinned API, described in §3; they are never pooled, and the word is used in this sense only |
| effort | the size of the reasoning budget a request asks the vendor for. Where a request names none the vendor's routing layer supplies its own default, which is the setting arm P-D switches on |
| quota | the vendor-side cap on requests per interval; a key that reaches it returns no completions until the interval resets |
| UNSCOREABLE, UNDERPOWERED, BELOW-FLOOR | a cell below its arm's registered floor of valid samples; a second-vendor chain arm with fewer than 5 scoreable cells; an arm V alias with fewer than 10 of 25 registered invocations valid. Each claims nothing |
| evaluability floor | the minimum valid-sample count a cell needs before its verdict counts, fixed at each arm's own registration and *not* common across arms; the per-arm values are in §3 |
| anchor | the template the proposer concentrates on at $N$, with its closed-form value. That value is the extended anchor $V(k^*, N - k^{*2})$ at a converge cell and the truncated anchor $T(k^*, N) = N/(2k^*)$ at a trap cell. At a discriminating cell it sits below the family argmax |
| $k$-inheritance | an output whose dominant circle radius gives the same grid order $k$ as its parent's; counted per parent condition in the arm MU table |
| located but not attributed | a difference measured between conditions differing in more than one way, so that the difference is established and its cause is not |

## 3 Registration, instruments, scoring, floors

**Registration protocol.** For each cell the prompt is pinned and its SHA-256 digest written to disk first. The $N = 13$ square prompt has a digest beginning `32db485b`, given in full in the supplement. Predictions live in the header of each arm's analysis script or in a standalone registration file. Competing predictions, numeric thresholds, tie conventions and a falsifier are fixed before the first invocation. Raw outputs are stored verbatim, failures included. Scoring is deterministic and local. Each arm writes its rows to a JSONL ledger. The supplement carries every arm's ledger, every departure from a registration, and the per-cell tables for the serving-path and second-vendor arms. Every reported verdict was registered before its arm's first scored run. Post hoc readings are labelled where they appear and listed together in §9.

**Two instruments, never pooled.** The first is an agent runtime. It does not expose decoding parameters, so every runtime row is a distributional claim over an observed sampling regime. It dispatches each call to a *subagent*, its worker process. That process inherits instruction files outside the task prompt. The digests do not lock those files, and a replicator cannot reconstruct them. The runtime also publishes no weight hash, so the alias-to-weights binding is a vendor promise. The second is a pinned API, the vendor's chat-completions endpoint reached through OpenRouter, at temperature 1.0 with no system prompt. It exposes and pins temperature and carries no inherited context. Arms MU and L run on the runtime. Arms P and P-D run on the pinned path. Arms GM, GM2 and GM3 use a second vendor's first-party API, and arm V uses free-tier aliases at seven attributed vendors. No count in this paper pools across instruments.

**Scoring conventions, fixed in advance.** Validity is reported at $10^{-9}$ and $10^{-6}$, both logged, with $10^{-6}$ primary. Proposers print six to eight decimals, so an eight-decimal tangency misses contact by about $5 \times 10^{-9}$. The primary threshold sits far below the gap of about $10^{-2}$ between the competing constructions a cell can express. Value matching uses a $2 \times 10^{-3}$ window. That window never carries a clearance verdict. A proposal clears the family argmax when its verified sum strictly exceeds the closed-form argmax at the tolerance named with the count. That argmax is the family's maximum by construction, so a clearance is an output outside the recipe family. Completions are parsed with `ast.literal_eval` after stripping code fences. One that fails to yield a list of $N$ strictly positive numeric triples is a logged parse failure, scored invalid. Every on-prediction rate is computed over valid outputs and so conditions on execution success. Per-cell validity denominators are reported beside every rate, except in arm MU. That arm registered three rates, the on-prediction rate (its registration calls it the anchor-rate), keep-or-improve and $k$-inheritance, pooled across its three cells. Only P-MU4 (§4) was registered per cell. Its table therefore gives pooled counts for those three and per-cell counts for P-MU4.

**Evaluability floors, per arm.** The evaluability floor is the minimum valid-sample count a cell needs before its verdict counts, fixed at each arm's own registration. The second-vendor chain uses 3: a cell below it is UNSCOREABLE for mode analysis and claims nothing in either direction. A chain arm with fewer than 5 scoreable cells is UNDERPOWERED. Arms P and P-D use 5. Arm V uses a per-model floor of at least 10 of a model's 25 registered invocations valid at $10^{-6}$. A model below it is labelled BELOW-FLOOR and carries no anchoring claim in either direction. Arm L's floor is a two-part rule on each lineage, the chain of generations grown from one cell (§5). A lineage needs at least 3 valid proposals at generation 0, the unconditioned start, and at least 1 in at least 3 of the 5 generations conditioned on a parent. All four lineages clear it. The weakest, lineage 13-GREEDY at $N = 13$, clears on 5 valid at generation 0 and on valid outputs in 3 of its 5 conditioned generations. Its 4 valid conditioned proposals are not the quantity the floor tests.

**The eight arms.** All eight were run once, in one registered corpus. Arms P and P-D answer an instrument question in the companion and a transfer question here, so both papers report the same rows in full (Anonymous authors, 2026). The other six are reported here and appear in the companion only as a pointer. Each arm's design is stated where the arm is reported. Supplement S1 tables all eight.

## 4 One conditioning step: arm MU

Arm MU hands the proposer one parent and one instruction, with no iteration. Given a parent at the family argmax the proposer keeps it, 26 of 26, and none of those outputs sits on the anchor. Given its own anchor as parent it leaves, 17 of 28 moving above it.

**Design.** Each call is the bare prompt, an existing packing, and the instruction to increase the sum of radii. The *anchor parent* is built at the model's own unconditioned anchor value $T(k^*, N)$. The *argmax parent* is built at the family argmax $V(k^*{-}1, m)$, which at these cells is provably better than the anchor. The *off-family parent* is a diagonal-chain packing scoring far below either. The registration and the supplement name them A_anchor, B_rival and C_offfamily. The cells are $N = 13$, 21 and 31. Three parent conditions in three cells, at $n = 15$ samples each, give 135 invocations.

**Registered predictions.** P-MU1 predicted an on-prediction rate under the anchor parent of at least 50%. P-MU2 predicted an on-prediction rate under the argmax parent of at least 35%. P-MU3 predicted $k$-inheritance under the argmax parent below 50%. P-MU4 predicted that under the off-family parent the modal valid output sits on the $k^*$-grid, scored per cell. The registered decision rule calls anchoring dissolved when the pooled on-prediction rate under the argmax parent is at most 20% and the keep-or-improve rate is at least 50%. The falsifier fires on the same condition.

**The rule's two conditions under the argmax parent.** Both are met, and they are not independent. The pooled on-prediction rate is 0 of 26 valid outputs. That meets the rule's 20% ceiling and fails P-MU2's 35% floor. Keep-or-improve is 26 of 26, above its floor of 50%. The argmax parent outscores the anchor $T(k^*, N)$ by 0.151 at $N = 13$. The gap is 0.159 at $N = 21$. It is 0.165 at $N = 31$. Each gap is more than seventy times the $2 \times 10^{-3}$ matching window, so an output that keeps or improves on that parent cannot sit on the anchor. The 26 of 26 entails the 0 of 26, and the rule reduces to one condition.

Keep-or-improve is almost entirely keep. Of the 26, 25 return the parent's score within $10^{-6}$, and one exceeds it by $1.2 \times 10^{-6}$. In coordinates, 18 reproduce the parent to within $10^{-6}$ on every coordinate. The $1.2 \times 10^{-6}$ excess is among them, from radii perturbed below that tolerance. One further output is a mirror image of the parent. The other 7 relocate filler circles while keeping the radius multiset and the parent's sum to seven decimals. All 26 inherit the parent's grid order too, so P-MU3 is disconfirmed.

The independent evidence is the anchor parent. Its on-prediction rate is 7 of 28, below P-MU1's 50% threshold, so the anchor is not self-stable as its own parent. Here the model departs: 17 of 28 move above the anchor and 4 fall below it. Grid order is kept by 22 of 28, 79% (60–90%). The 6 that change it all move above the anchor, and they include 3 that reach the family argmax value. The falsifier negates the persistence prediction over this pair of conditions, and no other was registered.

**The unconditioned rate on the same instrument.** Arm L's generation 0 (§5) supplies the no-parent reference on the same prompt bytes, channel and wrapper. It puts 4 of 9 valid outputs on the anchor at $N = 13$. It puts 7 of 8 there at $N = 31$. The anchor parent gives 4 of 12 and 2 of 11 at those cells. Arm L did not run at $N = 21$, so no unconditioned row exists there.

**P-MU4, held in two cells of three.** The anchor's grid order does reassert itself in the one condition registration predicted. Given the off-family diagonal parent, 20 of 34 valid samples abandon the parent and emit a $k^*$-grid construction. The prediction holds at $N = 21$ and at $N = 31$, and not at $N = 13$. The attractor captures the trajectory from a worse parent, and in 0 of 26 from a better in-family one.

| Parent | Valid | On-prediction (95% Wilson) | Registered threshold | Score against parent | $k$-inher. |
|---|---|---|---|---|---|
| None (arm L generation 0, same instrument) | 17 | 4/9 at $N = 13$ (44%, [19–73%]); 7/8 at $N = 31$ (88%, [53–98%]); no row at $N = 21$ | unconditioned reference, no bar | — | — |
| Argmax parent (a packing at the family argmax) | 26 | 0/26 (0%, [0–13%]) | dissolved if $\leq$ 20% | 26/26 (100%, [87–100%]; bar $\geq$ 50%): 25 equal within $10^{-6}$, 1 above by $1.2 \times 10^{-6}$ | 26/26 |
| Anchor parent (the model's own anchor) | 28 | 7/28 (25%, [13–43%]); per cell 4/12, 1/5 and 2/11 at $N = 13$, 21 and 31 | P-MU1 predicted $\geq$ 50%, not met | 17/28 above the anchor, 4/28 below | 22/28 |
| Off-family parent (diagonal chain, sum $\approx$ 0.6) | 34 | 20/34 abandon parent for $k^*$-grid (59%, [42–74%]); per cell 1/7 [3–51%], 7/13 [29–77%] and 12/14 [60–96%] at $N = 13$, 21 and 31 | P-MU4: $k^*$-grid reasserts, scored per cell | — | — |

Bracketed ranges are 95% Wilson intervals on the rate they follow.

**Clearance of the family argmax (post hoc, disclosed).** Arm MU's rows were scored against the anchor, so the clearance question was never asked at registration. Recomputing the argmax from the closed form gives 1.776142375 at $N = 13$. At $N = 21$ it is 2.258883476. At $N = 31$ it is 2.748528137. The 135 invocations give 88 valid outputs at the primary tolerance. At $10^{-9}$ just 63 of the 135 are valid. Eighteen of the 88 exceed the argmax at $10^{-6}$. Seventeen of the eighteen are argmax-parent outputs returning their parent's score within $10^{-6}$. The eighteenth is the anchor-parent output at $N = 21$ that moved onto the argmax.

The argmax parent's seven-decimal coordinates sum to $3.25 \times 10^{-7}$ above the closed form at $N = 13$. At $N = 31$ that gap is $2.63 \times 10^{-7}$. Each also fails the $10^{-9}$ test on one overlapping pair of circles, not on containment. The amount by which each of the eighteen exceeds the argmax equals the overlap on its one failing pair, which the $10^{-6}$ tolerance permits. Their largest pairwise overlaps run from $3.8 \times 10^{-9}$ to $4.3 \times 10^{-7}$. The median is $4.4 \times 10^{-8}$. The eighteen sit above it by more than the $5 \times 10^{-8}$ granularity at which their sums are recorded. Nine further rows of the 88 exceed it only within that granularity. The remaining 61 of the 88 do not. At $10^{-9}$ none of the 63 clears.

## 5 Five loop generations: arm L

Arm L iterates arm MU's conditioning step for five generations, feeding each scored proposal back as the next parent. Iteration keeps the proposer off the anchor: 11 of 49 conditioned outputs sit on it, against 11 of 17 at generation 0. Three of the four lineages end exactly on the family argmax and none goes above it.

**Design.** Both prompts were hashed, and the stepper and scorer were committed alongside. Generation 0 is the bare prompt, sampled fresh inside this arm. Its prompt is byte-identical to the companion paper's unconditioned arms, served on the same channel and wrapper. Generations 1 to 5 use arm MU's registered mutation prompt verbatim, instantiated with the parent packing and its score. Two discriminating cells, $N = 13$ and $N = 31$, are crossed with two archive regimes. GREEDY retains only the single best valid proposal. DIVERSE retains up to five distinct-valued ones. That gives four lineages of six rounds at population 5, which is 120 invocations. Two of the 120 were rejected by the runtime before reaching the model, and are counted as invalid rather than dropped. The parent for population slot $i$ is archive[$i$ mod |archive|], so the loop carries no sampling randomness of its own. All four lineages are evaluable under the registered floor.

This is a minimal analogue. It has no islands, no crossover and no program representation. Its only evaluator signal is the parent's own score.

**Registered predictions.** L1 predicted that the modal valid output at generation 0 equals the anchor in every lineage. L2 predicted that dissolution survives iteration, so that the pooled on-prediction rate over generations 1 to 5 falls below the generation-0 rate. L3 predicted that the pooled conditioned on-prediction rate is higher under GREEDY than under DIVERSE, on the reading that loop diversity is scaffold-produced. L4 was registered open with no direction predicted: report best-of-run against the family argmax. The falsifier fires if the pooled conditioned rate meets or exceeds the generation-0 rate in both cells.

Table 3: Arm L by lineage. Valid outputs over invocations, and on-prediction counts, at generation 0 and across the five conditioned generations, with each lineage's best scoring valid output, the generation at which that best first appeared, and its cell's family argmax. Pooled: 11 of 17 on-prediction at generation 0, 65% (95% Wilson 41–83%), against 11 of 49 conditioned, 22% (13–36%).

| Lineage | Gen-0 valid / invocations | Gen-0 on-prediction | Conditioned valid / invocations | Conditioned on-prediction | Best of run | Best first reached (gen) | Family argmax |
|---|---|---|---|---|---|---|---|
| 13-GREEDY | 5/5 | 2 | 4/25 | 0 | 1.7761424 | 0 | 1.7761424 |
| 13-DIVERSE | 4/5 | 2 | 16/25 | 5 | 1.7761424 | 5 | 1.7761424 |
| 31-GREEDY | 4/5 | 3 | 18/25 | 3 | 2.6104167 | 5 | 2.7485281 |
| 31-DIVERSE | 4/5 | 4 | 11/25 | 3 | 2.7485281 | 2 | 2.7485281 |
| Pooled | 17/20 | 11 | 49/100 | 11 | | | |

**Validity, and what every rate in this arm conditions on.** Validity drops under iteration. The pooled row of Table 3 gives the two totals. The weakest lineage, 13-GREEDY, carries the loss. Conditioned rounds lose validity as well as accuracy, so every rate here counts only survivors. The 11 of 49 on-prediction rate is one such rate. Over all invocations it is 11 of 100. That is 6–19%. At generation 0 it was 11 of 20. That is 34–74%. The registration set no validity prediction, so the loss is a post hoc observation.

**L2 confirmed, with unequal support.** Sum the two lineages at each cell of Table 3. At $N = 31$ the generation-0 on-prediction rate is 7 of 8. Conditioned, it falls to 6 of 29. That is 21% (10–38%). The two Wilson intervals there are disjoint. At $N = 13$ it starts at 4 of 9. Conditioned, it is 5 of 20. That is 25% (11–47%). There the intervals overlap and the drop is a point estimate. The falsifier did not fire. Once a lineage's parent sits off the anchor, later rounds condition on a parent at the family argmax, arm MU's second condition, so L2 is not independent of MU. 13-GREEDY returned 4 of 25 valid under that condition where arm MU's argmax-parent rows returned 26 of 45, a gap this paper does not explain.

**L1 was not met.** It fails on one lineage. The 13-DIVERSE seed round in Table 3 splits evenly, and the registered tie rule counts the tie against the prediction. The other three lineages pass. The rule set four per-lineage bars rather than one pooled bar per cell. Each bar ran over the four or five valid proposals that lineage's seed round returned.

**L3 was not met.** Pool each archive regime over both cells. The conditioned on-prediction rate is 3 of 22 under GREEDY. That is 14% (5–33%). Under DIVERSE it is 8 of 27. That is 30% (16–48%). The difference runs opposite to the registered direction and is not separable from chance. Fisher's exact test on the $2 \times 2$ table gives $p = 0.303$. The odds ratio is 0.375. The comparison is also confounded: lineages differ in parent identity as well as archive width. 13-GREEDY also contributes the fewest conditioned valid proposals of any lineage. The outcome is a disconfirmation of L3, not evidence for its converse.

**L4: three of the four lineages reach the family argmax exactly, and none exceeds it.** The best-of-run and family argmax columns of Table 3 carry the comparison. Both lineages at $N = 13$ and 31-DIVERSE land exactly on their cell's family argmax, and 31-GREEDY stops short of it. The best of 13-GREEDY is a generation-0 output, one family-argmax packing in its unconditioned seed round that the loop never bettered. The other three lineages reach their best under conditioning. No valid conditioned output the table counts exceeds the best value the recipe family can express at $10^{-9}$, the clearance tolerance. One post-sampling scorer correction, affecting only that $10^{-9}$ test and no verdict, is disclosed in Supplement S2.

## 6 Serving path and reasoning budget: arms P and P-D

Arms P and P-D change how the proposer is served, not what it is conditioned on. On the bare pinned path validity collapses, 0 of 105 at the square cells, so no mode prediction can be scored. With extended thinking on, validity returns, 12 of 14, and the anchor is again the plurality value. Per-cell tables sit in Supplement S5.

**Arm P: the registered prompt on a bare, pinned serving path.** Arm P sends the registered prompts byte-identical through a path that exposes and pins temperature. It runs at $n = 15$ per cell across seven square cells, five held-out cells and three code cells. That is 225 invocations. The serving path refused 6, none at a square cell, each counted invalid where it was issued. Two competing mode predictions say the anchor $T(k^*, N)$ is modal in a majority of the seven square cells, or in few of them. Two competing code-channel predictions say no valid program clears the family argmax, or at least one does. The falsifier is on the discriminating-cell rate. Supplement S1 states all five in the registration's own words. The registration left 4 of the 7 square cells that neither mode prediction adjudicates, a design gap. An outcome there confirms neither.

**Validity collapsed.** The seven square cells carry both mode predictions. There, 0 of 105 outputs are valid at either tolerance. All 105 returned well-formed lists of the requested count, with no prose, fences or reasoning tokens. Every failure but three is an overlap. Supplement S5 names the three. The median maximum pairwise overlap at $N = 13$ is 0.10. That is fifty times the value window. At the held-out cells 4 of 75 are valid, all four at $N = 50$. They sum to 2.5, a full unit below the family argmax there, so none clears.

**Against the registered floor.** The arm's own floor is five valid outputs per cell, so the direct-emission predictions are UNSCOREABLE at every cell. The falsifier cannot fire either. Its denominator, the valid outputs at the four discriminating square cells, is empty. We report the empty denominator as empty, and impute no rate to it. On the code cells 12 of 45 programs are valid, and only $N = 21$ reaches the floor. None clears the family, so the no-clearance prediction holds.

**Two post hoc diagnostics bound the cause.** Neither was registered. The first is a temperature sweep. At 0.0, 0.5 and 1.0 the $N = 13$ prompt gives 0 of 6 valid at each. The second is a same-day control at the same cell through the agent runtime, tabled beside the two arm P-D conditions in Supplement S5.

**Arm P-D: which component of the instrument.** Arm P-D was registered after an outside review asked for a diagnostic beyond temperature. It sends the same $N = 13$ prompt, to the same alias, on the same pinned path and decoding, under two one-line manipulations, at $n = 15$ each. The *thinking-on* condition turns the vendor's extended thinking on at the routing layer's default effort and sends no system prompt. The *thinking-off* condition leaves thinking off and adds one system line asking for care. The supplement labels them D1 and D2. The reasoning-budget hypothesis predicts thinking-on valid in at least 8 of 15 against at most 3 for thinking-off. The context hypothesis exchanges the two thresholds. A third registered outcome covers anything else. The arm closed at 27 of 30 rows by a registered amendment, after the serving path refused the last three for credit. Those rows could not have changed the verdict, because thinking-off could reach at most 2 valid while thinking-on already held 12. Thinking-on is valid in 12 of 14. The 95% Wilson interval on that rate is 60–96%. Thinking-off is valid in 0 of 13, every failure an overlap, exactly arm P's collapse. So the reasoning-budget hypothesis holds, and the anchor is the plurality value. $T(4, 13) = 1.625$ accounts for 5 of the 12 valid thinking-on outputs. The registered secondary prediction, that this value is modal in whichever condition clears the validity threshold, is therefore met. Only $N = 13$ was run under the two conditions, so the component is named on one cell, and no interval is computed there.

## 7 Other vendors: arms GM, GM2, GM3 and V

Arms GM, GM2, GM3 and V change the vendor. GM3, the only chain arm with a verdict, meets the pooled bar, 46 of 80, and misses at the discriminating cells, 11 of 43, where the modal output is the family argmax construction. In arm V ten of thirteen aliases fall below the floor, and the three that clear it split, two anchoring at point estimate and one not. Arms GM, GM2 and GM3 run against a second vendor's first-party API and vary the model family, with prompts pinned, an output budget fixed and a provider logged per row. Arm V varies vendor and serving class at once. The per-cell tables and the chain's verdict table are in Supplement S4.

**Arms GM and GM2 returned no scoreable data, and both are reported.** One chain registration fixes the definitions and predictions for all three GM arms. Each arm addresses seven $N$ cells by twenty samples, over a floor of three valid samples per cell. The primary prediction requires a mode match in at least 5 scoreable cells. The second requires a pooled on-prediction rate of at least 30%. The third requires the two-radius signature at order $k^*$ in at least half of on-prediction samples. That signature is a grid radius $1/(2k^*)$ carrying a filler radius $(\sqrt{2} - 1)/(2k^*)$. The falsifier fires on a mode-match failure in at least 4 scoreable cells. Arm GM addressed a gemini-flash-lite alias, part of it through a routed path (Supplement S2). It returned 20 valid of 140, UNDERPOWERED under both accountings, and its falsifier fires under neither. Arm GM2 addressed the gemma family at a 4,096-token output budget, and returned 0 of 140 parseable. The cause is a budget confound, not a model result. The vendor API returns deliberation on a separate channel, and the budget was spent there before any answer text. Arm GM3 re-ran the same design at 16,384 tokens. Of its 140 rows, 17 did not parse. Of the 123 that did, 80 are valid packings at $10^{-6}$.

**Arm GM3: the pooled bar is met and the discriminating cells are not.** Arm GM3 runs the unconditioned-call protocol against gemma-4-26b through the vendor's first-party API. The first-party key reached its quota, so 41 of the 140 rows were sampled through a routed path under an amendment committed before resumption, and both accountings are reported.

The pooled prediction held. Of the 80 valid packings, 46 land within $2 \times 10^{-3}$ of the value the closed form predicts. That clears the registered pooled bar. The 41 routed rows carry 2 of the 80 valid packings. None of the 46 on-prediction are among them. First-party rows alone therefore give 46 of 78, and no verdict changes. A pooled rate cannot separate family search from anchoring, because the two agree off the discriminating cells. The two cells where the anchor equals the family argmax hold 35 of the 46 on-prediction packings. On the three scoreable discriminating cells the rate is 11 of 43, below the bar the pooled metric passes.

The cell-level bar was not met. The anchor is modal in 2 of 5 scoreable cells, and Figure 1 plots the per-cell modal sums against the anchor. The falsifier did not fire either, because it needed failure in at least 4. Bar and falsifier together leave one to three failures unadjudicated, a registration design gap, and the observed three fall in it. The two-radius signature also came in under its bar of half. Only 20 of 46 on-prediction samples carry the exact decomposition. That bar cannot be met at a truncate cell, whose anchor carries one radius and no filler. The 15 on-prediction samples at $N = 35$ are counted against a signature they could not show.

**The misses are the family argmax construction (post hoc, disclosed).** The modal output at every discriminating cell is the family argmax itself. A post hoc structural diagnostic finds all 27 of the 27 packings at the family argmax value carrying its exact two-radius decomposition. That decomposition is the $(k^*-1)$ grid radius with the $(k^*-1)$ filler radius. The registered signature test could not see this, because it tests

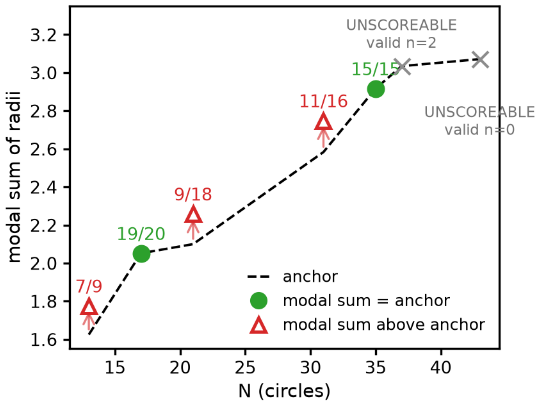


Figure 1: *Arm GM3 modal sums against the anchor.* Filled green circles match the anchor (the dashed line is $V(k^*, m)$ at extend cells and $T(k^*, N)$ at truncate cells), with frequencies annotated. Open red triangles miss, and every miss lands above the anchor at the family argmax *value*. The post hoc structural check in the text, not this figure, establishes that these are also the family argmax *construction*. Grey crosses on the anchor line are the two UNSCOREABLE cells, with their valid counts. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json`.

radii against the predicted order $k^*$. On the discriminating cells 27 of 43 valid packings are the family argmax construction. That is 63%, on a 48–76% interval. Validity collapses at the two largest cells, $N = 37$ and 43, which bounds where this model returns evaluable packings. Both cells are UNSCOREABLE under the registered rule and claim nothing. Clearance was re-read post hoc, as for arm MU. Of the 80 valid packings, 79 are valid at $10^{-9}$. None clears the family argmax at either tolerance (Supplement S4).

**Arm V: thirteen free-tier aliases.** Arm V sends the GM chain's prompt files byte-identical at five cells, $N = 13$, 17, 31, 35 and 37. It draws five samples per model and cell. The targets are free-tier aliases across seven attributed vendors. Three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths. The last raised the output budget after hidden-reasoning truncation emptied three models' replies.

Execution retried each invocation slot until content arrived, per the registered transport-repair rule. The per-model floor of §3 is evaluated on registered slots, so a slot with no row counts against it. Ten of thirteen aliases finished BELOW-FLOOR, including all three originally registered aliases. One of them made 108 attempts across its 25 registered slots, none of them with content. The other three cleared the floor, and their registered verdicts split.

V1 predicted at least 50% of a model's valid outputs on-prediction, and V2 at most 10% at the family argmax. V2 is scored only at the discriminating cells, $N = 13$ and 31. In this arm, *transfer* denotes only the registered pooled-bar verdict. An alias must clear the per-model floor and meet the V1 bar on its valid outputs pooled over five cells. Every transfer verdict here rests on amendment-added aliases. All three were registered by amendment 2 (`arm_v_preregistration_amendment2.md`, `d579797`).

| alias | vendor | valid | V1 on-prediction | V1 bar ($\geq$50%) | V2 family argmax valid at discriminating cells (bar: $\leq$10%) |
|---|---|---|---|---|---|
| `north-mini-code` | Cohere | 12 | 8/12 [39–86%] | **MET** (point est.) | 0/4 **HOLDS** |
| `gpt-oss-20b` | OpenAI | 18 | 11/18 [39–80%] | **MET** (point est.) | 0/9 **HOLDS** |
| `gemma-4-31b-it` | Google | 15 | 0/15 = 0% [0–20%] | **NOT MET** | 0/3 **HOLDS** |

Bracketed ranges are 95% Wilson intervals on the rate they follow. V2 is scored only at the discriminating cells. The registration labels a met V1 bar TRANSFERS and a missed one DOES-NOT-TRANSFER; this table reports the bar outcome, in the sense §7 defines. Both passing intervals span the 50% bar.

**Both passes are weaker than the percentages suggest.** The bracketed Wilson 95% intervals include the bar, so each verdict is a point-estimate pass. And the pooled rate spans cells that cannot discriminate anchoring from family search. Decomposing onto the discriminating cells gives 3 of 4 for the first alias. It gives 5 of 9 for the second. The arm's strongest reading, that these aliases behave as the original family does, does not survive the template-shape null. That null is a proposer picking uniformly at random among the three grid orders $k^*-1$, $k^*$ and $k^*+1$, one branch value each. It lands on-prediction one time in three. Supplement S5 derives it in this paper's own terms and gives the exact tails, and this paper applies it post hoc (Anonymous authors, 2026). The pooled rates clear that null only because most of their cells cannot discriminate. At the cells that can, neither pass separates from it. Arm GM3's 11 of 43 at its discriminating cells does not separate from it either. V2 is a point-estimate pass on 16 valid outputs at discriminating cells: 0 land on the family argmax, on per-alias counts of 3, 4 and 9 whose Wilson intervals all include the 10% bar. The same post hoc re-read over the three evaluable aliases finds 45 valid at $10^{-6}$ and 39 at $10^{-9}$. None clears the argmax. The closest falls short of it by $2.5 \times 10^{-3}$ at $N = 31$ (Supplement S4).

**Anchoring is not confined to the original family, and not universal.** Two further vendors' models land the registered construction values at point estimates above the bar. Under the chain's first-party instrument, gemma-4-26b emits the family argmax construction at the discriminating cells. Under arm V's free-tier instrument, gemma-4-31b lands neither the anchor nor the family argmax. Its valid packings are de-novo, unanchored and weak. Their per-cell bests run 0.88–1.85. The anchors at those cells run 1.63–3.03. Model variant and instrument change together across this pair, so the boundary is located but not attributed. Either way, no single-model measurement would have seen it.

## 8 Related work

**The loop the proposer is characterized for.** Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt (Guo et al., 2024), EoH (Liu et al., 2024) and LLaMEA (van Stein & Bäck, 2025) made the LLM call an evolutionary operator over prompts and programs. FunSearch (Romera-Paredes et al., 2024) turned the pattern into a discovery claim. AlphaEvolve (Novikov et al., 2025) and its successors inherit that claim, OpenEvolve (Sharma, 2025) among them (Lange et al., 2025; Su et al., 2026; Khrulkov et al., 2025; Cemri et al., 2026; Wang et al., 2025). Every one conditions its proposer on a parent and iterates it, two of the conditions this paper varies. Simple baselines (Gideoni et al., 2026) and classical solvers (Berthold et al., 2026) recover much of the reported advantage.

**Template convergence under iteration.** "Mutation Without Variation" (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates. In 87% of chains, over 93% of mutations revisit a seen form. It runs mutation loops in program space with no closed form and no preregistration, and is the closest published neighbour to arm L. Its result is about structural revisiting. Arm L's second prediction is about a value the revisited structures do not cross.

**Anchoring, and what kind of object anchors here.** Anchoring is studied as a general bias over initial information, and as numeric primes dragging point estimates (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025). The modality here differs: a construction template has a computable value, where a scalar anchor has only a measurable pull. One level up, LLM outputs concentrate where human ones do not (Chen et al., 2026; Jiang et al., 2025). Set-level and diversity-aware selection (Li, 2026; Wang et al., 2026; Yang et al., 2026) and the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025) converge on the same question about the proposer's contribution.

**Two results that cut against this paper's framing.** Evolutionary search matters empirically (Zhang et al., 2024), so one unconditioned call is a poor stand-in for a loop. Strong LLM optimizers act as local refiners (Zhang et al., 2026b), as a parent-conditioned call does.

**Instrument effects and reasoning budget.** Larger and more instructable models become less reliable (Zhou et al., 2024). A reasoning model can report a higher hallucination rate alongside higher accuracy (OpenAI, 2025), and reasoning collapses past a complexity threshold (Shojaee et al., 2025). Geometric construction is a regime where nominal capability and executed correctness diverge (Kim et al., 2026; Balunović et al., 2025).

**Preregistration lineage.** Recipes, outcome-blinded predictions and agent protocols are already preregistered (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026). HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. This paper's arms lock a numeric threshold and a falsifier before each setting was changed.

## 9 Limitations, reproducibility, deviations

**Scale.** Arm MU takes one conditioning step, and arm L runs five generations at population 5 with one mutation operator. A larger loop, with islands and crossover, could sustain variation this one cannot, so scale stays open. The two contrary results in §8 are consistent with what this loop shows.

**No instruction-free control.** Every conditioned call asks for a larger sum, and no call without that instruction was run. So returning the argmax parent unchanged, 26 of 26, cannot be split into recognising an optimum and ignoring the instruction. Only its parent-dependence is established: the same instruction moves 17 of 28 off the model's own anchor.

**Vendor and tier scope.** The proposer under test is one vendor's weak-tier alias, except where another is named. The cross-vendor arms move the pooled-level statement, not the cell-level mode structure.

**Prompt robustness.** Every arm conditions on a single hash-locked prompt. That buys reproducibility, not robustness to rewording. The companion paper reworded two discriminating cells and found the same modal construction (Anonymous authors, 2026).

**Untested alternatives.** Typicality bias in preference-tuned models (Zhang et al., 2026a) predicts concentration on the most typical construction without reference to geometry, and nothing here excludes it. Contamination probes (Carlini et al., 2021; Golchin & Surdeanu, 2024), a format-sensitivity sweep (Sclar et al., 2024) and alignment-induced diversity loss (Kirk et al., 2024) remain untested.

**Reproducibility.** Every arm's prompts were hashed, and its predictions and falsifiers fixed, before sampling. The exceptions are the ones the supplement's deviation table records, and no arm was selected on its result. The anonymized artifact holds the ledgers, scripts, registrations, amendments and frozen reports. It regenerates every number by a documented command. Arms P and P-D can be re-sampled on their pinned path. Arms MU and L can be re-scored but not re-drawn, because their rows depend on a dispatch wrapper and inherited context the digests do not lock.

**Where the forking paths are.** Five choices were made with data already in hand, and each is named here. (a) Arm V's evaluable set exists because of amendment 2 (`d579797`), which expanded the alias list after the three registered aliases finished below the floor. (b) Arm P-D was designed after arm P collapsed, and registered at `5554cc0`. (c) The clearance readings for arms MU, GM3 and V are post hoc re-reads, registered for none of them. (d) Both arm P diagnostics, the temperature sweep and the same-day runtime control, were chosen after that collapse. (e) Arm P's scoring code was committed at `596b113`, after its registration at `aa473ad` and after sampling. None is corrected for multiplicity. The protection is that each is listed.

**Deviations, in summary.** Full rows are in the supplement. Arms GM and GM3 each carry one registered serving-path deviation, with both accountings reported for each, and GM3 also carries a disclosed duplicate-row repair. Arm V has three pre-sampling amendments. Arm L has one post-sampling scorer correction, affecting only the $10^{-9}$ clearance test. Arm P-D closed at 27 of 30 rows by an amendment that could not change its verdict.

**Use of AI systems.** Claude models drafted prose and assisted implementation. The authors designed the study, specified the instruments and their decision rules, and approved each preregistration before sampling. Each such commit is a git ancestor of the sampling it governs.

## 10 Conclusion

Eight registered arms changed four things about the setting, for one task and one weak-tier proposer. On the agent-runtime instrument, one conditioning step under the improve instruction left 0 of 26 outputs on the anchor given a better parent. An off-family parent brought the $k^*$-grid back in 20 of 34. Five loop generations there did not restore the anchor. A pinned serving path collapsed validity until extended thinking was on. At a second vendor the pooled bar held while the discriminating cells did not. Table 1 carries the rest. A zero-shot characterization therefore licenses no prediction about a proposer inside a discovery loop without a measurement at that loop's own conditions. The boundary is two-sided: the anchor does not carry into a conditioned call, the family argmax does, as a ceiling no arm clears.

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